

Multiple Choice (10 marks)

- Which of the following is incorrectly named?
 - MgO magnesium oxide
 - AgNO₃ silver nitrate
 - CuCl₂ copper(II) chloride
 - H₂SO₄ sulfuric acid
 - K₂Cr₂O₇ potassium dichromate
- Which of the following geometries corresponds to a substance that has five sigma bonds and one nonbonding pair of electrons?
 - Bent
 - Trigonal pyramid
 - T-shaped
 - Trigonal planar
 - Trigonal bipyramidal
- What is the mass of one molecule of cholesterol (C₂₇H₄₆O, molecular mass = 386)?
 - $3 \times 10^{-23}g$
 - $7.15 \times 10^{-22}g$
 - $1.38 \times 10^{-21}g$
 - $6.41 \times 10^{-22}g$
 - $8.41 \times 10^{-22}g$
- Solve $6x^2 - 7x - 20 = 0$.
 - $5/2, 4/3$
 - $-5/3, 4/2$
 - $5/3, -4/2$
 - 2, -10
 - 4, -5
- Calculate the pressure exerted by 1.0 molXe when it is confined to 1.0dm³ at 25°C.

- (a) 21atm
- (b) 28 atm
- (c) 30 atm
- (d) 12 atm
- (e) 18 atm

True / False (10 marks)

Answer True or False

- 6. True or False: The resulting molarity of the sucrose solution after mixing is 0.45 M.
- 7. True or False: The nuclide mass of N^{13} is accurately represented as 13.00335 amu.
- 8. True or False: The pH of a 0.001 M aqueous solution of KOH is 13, indicating a strongly basic solution.
- 9. True or False: The work function of potassium metal is $4.65 \times 10^{-19}J$, corresponding to its threshold wavelength.
- 9. True or False: The volume of O_2 necessary to completely burn 50 liters of CO is 75 liters, producing 150 liters of CO_2 .

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Calculate the pH of a 0.2 M NH_3 solution given $K_b = 1.8 \times 10^{-5}$ at 25°C. Discuss the implications of your findings.
- 11. Discuss the chemical reaction that occurs when Fehling solution is added to urine containing sugar from a diabetic. What red copper compound is formed upon heating, and what does this indicate about the urine's composition?

End of Paper